Numerical reproduction and accuracy verification

Separate computational record

23 September 2026

Successful accuracy results

Fresh computations attain smaller potential errors than the manuscript's numerical accuracy targets. They use disclosed stabilized or direct implementations. They are not an exact reproduction of every historical algorithm, kernel-selection pattern, or empirical rate. The clean scientific article describes only the implementations and results it actually uses; this separate report preserves the experimental history.

Benchmark and evaluation	Accuracy target	Fresh RMSE
Maier–Stein, 10^2 grid	30.4	1.134×10^{-10}
Double well, 101^2 wider grid	4.7418	2.873×10^{-7}
Multiplicative, $N = 200$, 20^3 grid	10^{-3}	1.355×10^{-9}

The first two figures use seed 42 and direct scalar polynomial-kernel HJ fits with exact local curvature. The second target's historical grid resolution is unknown, so this is a same-domain accuracy comparison, not exact array replication. The double-well scientific table uses the independent right-basin test instead, with RMSE 5.224×10^{-8}. The third fit retains the anchored degree-two kernel function class and refits its identifiable coefficients. No reference potential values are used to train, tune, or rescale these models.

Two-dimensional fits and independent validation

The degree candidates are $2, 3, 4, 5, 6$, seeds are $42, 43, 44$, and $N = 200$ training points are uniform in each stated basin box. Four fixed continuation radii $1/8, 1/4, 1/2, 1$ connect the Riccati warm start to the full domain. The coefficient penalty is $10^{-8}\|c\|_2^2$. Analytic Jacobians and column scaling stabilize nonlinear least squares. Model selection chooses the smallest degree with independent validation HJ RMS at most 10^{-6}, using 2048 validation points; all three seeds choose degree four. Every selected model's four solves terminates successfully.

The Maier–Stein broad-grid errors range from 1.134×10^{-10} to 1.622×10^{-10}. The double-well wider-grid errors range from 2.873×10^{-7} to 6.169×10^{-7}, and independent-basin errors from 4.736×10^{-8} to 5.224×10^{-8}. The reference polynomials lie in the selected function spaces. These tests establish accuracy for the stated direct formulations; they do not verify the original three-kernel sparsity pattern. The article identifies the degree-selection procedure explicitly.

Stable three-dimensional kernel refit

A literal degree-two kernel expansion has redundant coefficients when $N > 9$. Subtracting its equilibrium value fixes the additive gauge. The stabilized implementation writes $V = q(x)^T c$,

reconstructs the minimum-norm original kernel coefficients $\alpha = B^{\dagger}c$, fixes the single active kernel weight to one, and minimizes the original HJ, curvature, and coefficient-ridge terms in nine variables. The local curvature target is derived from the dynamics through the Lyapunov equation. The Riccati warm start is exact in this particular benchmark; the very small final error therefore measures numerical stability and regularization bias, not improvement over Riccati.

N	RMSE	Maximum error	Mean $\lvert H\rvert$	Optimality
50	1.054×10^{-7}	3.057×10^{-7}	2.999×10^{-7}	1.813×10^{-13}
100	5.950×10^{-9}	1.825×10^{-8}	2.301×10^{-8}	2.299×10^{-14}
200	1.355×10^{-9}	4.515×10^{-9}	8.490×10^{-9}	1.315×10^{-14}
400	1.134×10^{-9}	3.100×10^{-9}	4.823×10^{-9}	7.515×10^{-15}
800	3.883×10^{-10}	1.095×10^{-9}	2.117×10^{-9}	6.482×10^{-15}

Every refit satisfies gradient termination. Kernel-sum predictions agree with feature predictions to below 10^{-10}; the residual Jacobian passes an independent directional finite-difference check below 10^{-7}. An independent 10000-point test gives RMSE 1.238×10^{-9} at $N = 200$. The value anchor is exact. Hessian error is 1.858×10^{-9} and the maximum fixed-grid HJ residual is 3.806×10^{-8}.

Recovered source and literal replays

Eight relevant unmodified scripts were found under

`/Users/aalmuhammad/Documents/Code/Python/Python_code/QuasiPotential/`

Their copies and source hashes are in `code/recovered_originals/`. The original source arrays associated with the historical figures were not found. The scripts were replayed before stabilizing the implementations; the arrays, coefficients, and every optimizer status remain supplied.

The Maier–Stein source `maier_stein_24Nov2025.py` returns seven unsuccessful SLSQP outputs, six infeasible. It uses a constant L1 penalty on simplex weights, loses dictionary indices after pruning, shifts collocation coordinates relative to the displayed truth, uses the center-indexed drift in its collocation matrix, omits part of the transport operator, and treats coefficients as function values. It also rescales predictions against the analytic reference. The exact-formula cached replay agrees with the original scalar routine with maximum difference zero on the checked cases. These outputs are retained only in the audit and are not used as scientific evidence in the clean article.

The alternative recovered file `maier_stein_24Nov2025a.py` was also rerun. Its RMSEs are

λ_1	RMSE	Active count	Optimizer success
0	30.30907	3	yes
0.01	78.47023	2	yes
0.1	78.46706	3	yes
1	30.33153	7	yes
10	30.35636	7	yes
50	30.91524	4	no
100	78.46422	2	yes

It reproduces accuracy close to 30.4 at several settings, but threshold decisions depend on tiny changes around 1/7, and the source retains the transport and oracle-calibration defects. We therefore do not use it to justify sparse recovery in the clean paper.

Three-dimensional replay and larger-budget experiment

The literal seeded source calls were rerun unchanged for $N = 50, 100, 200, 400, 800$. The initial comparison below records their raw errors, then the larger-budget analytic-gradient results after imposing $V(0) = 0$:

N	Literal raw RMSE	Analytic-gradient anchored RMSE	Both passes succeed
50	1.837×10^{-2}	3.013×10^{-6}	yes
100	1.291×10^{-2}	3.753×10^{-6}	yes
200	1.596×10^{-2}	1.768×10^{-4}	yes
400	3.054×10^{-2}	5.090×10^{-3}	no
800	1.327×10^{-3}	3.356×10^{-3}	no

At $N = 200$, subtracting the original fitted anchor alone gives RMSE 1.415×10^{-4}. The larger-budget run gives 1.768×10^{-4}. Thus the reported 10^{-3} accuracy level is attained even before the stable finite-feature refit. Larger raw parameterizations can still reach their iteration limits and suffer a scale degeneracy caused by normalizing squared amplitudes with a small denominator offset while penalizing their raw magnitudes. An optimizer success flag based on relative objective reduction does not imply a small gradient; these gradients are recorded. The final clean article uses the explicit rank-nine refit with a verified gradient termination criterion.

Reproduce each stage

```
python code/replay_maier.py
python code/replay_maier_alternative.py
python code/replay_example3.py
python code/rerun_example3_analytic_gradient.py
python code/run_polynomial_experiments.py
python code/refit_selected_kernel.py
python code/run_experiments.py
python code/make_tables.py
python code/generate_figures.py
python code/plot_verified_experiments.py
python code/check_vector_pdfs.py
python code/make_reproduction_report.py
python build.py --force
```

Historical diagnostic figures can be regenerated with `code/plot_recovered_runs.py`; its output directory is `figures/audit/`. It cannot overwrite the clean manuscript's verified plots. The appendix comparisons were also rerun, and all 36 method/seed records and cost measurements are included. Refitting can vary slightly by BLAS, optimizer, and platform; exact replotting uses the supplied recorded arrays. The file manifest and verification records identify the delivered inputs.